Prototype Design Layout:

Controller: ESP32

- Contains Wifi connection
- Maintains autonomous controls

Motor Requirements:

- 2 Brushless Motors to power Coaxial propellors
- 2 Servo Motors to tilt rotor cage

Drive requirements:

- 2 Electronic Speed Controllers (ESC) to control the 3 phases of the brushless motors
- Servos can directly connect to the ESP32, but need a separate battery source

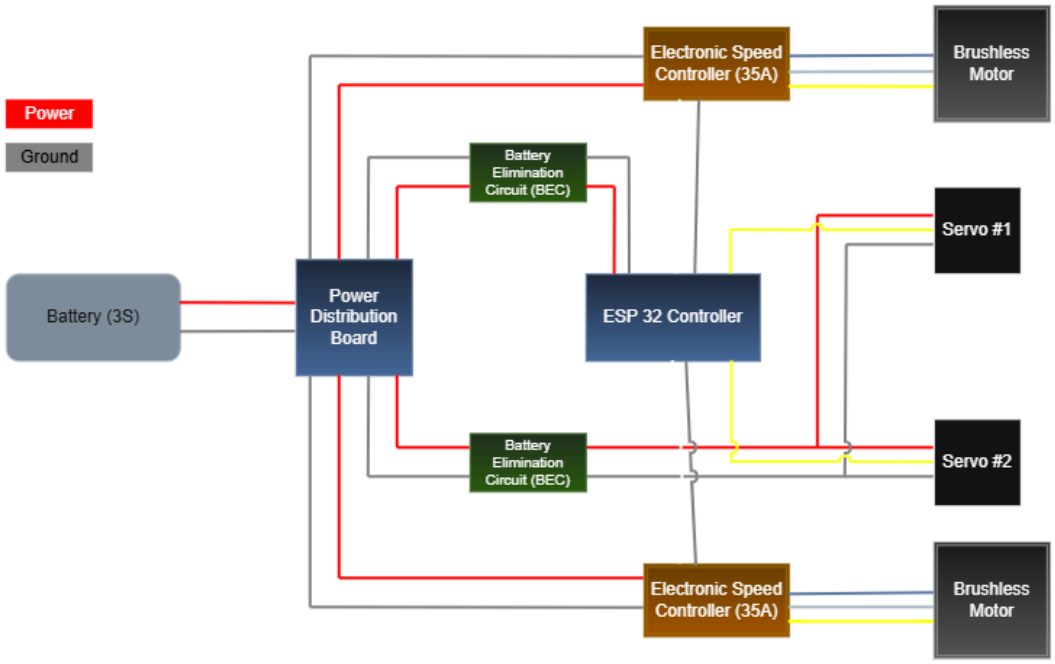
Sensor Requirements:

- IMU
    - One Accelerometers per direction, required for controlled landing
- GPS (Optional)
- Airspeed Controller
- SBUS receiver (Maybe?)

System Power Requirements:

Components	Operating Voltage/Amperage	Power Requirements
Battery (3S) • 3S = 11.1V • 3500 mAh	11.1V 3500mAh = 35A for 6 minutes	
Power Distribution Board • Separates power in parallel (same Volt, diff current)	11.1V across 6 outputs + 5V + 12V outputs 5V can be connected to the servos	
2x Brushless Motors • CR3516-1030 (Himax) • Rec. Battery: 3S • ESC rating: 35A		800 W
2x Electronic Speed Controllers (35A) • NiDiCi 35A	7.4V(2S) - 18.5V(5S) With 3S: 11.1V 35A	

• Compatible 2S-5S		
2x Servo Motors • kst x08 plus	3.8V-8.4V Max Torque = 5.3 Kgfcm @ 8.4V	
ESP 32 • USB-C Connection	3.3V	
2x Battery Elimination Circuit • Drops voltage into ESP32 and Servos • QWinOut UBEC-8A 2S-6S 6-36V		
Inertial Momentum Unit (IMU) •		



Important Links

Himax: https://www.buddyrc.com/collections/himax-contra-motors

2x UBEC: https://qwinout.com/collections/ubec/products/ubec-8a-2s-6s-6-36v-bec-dual-ubec-8a-16a-5-2-6-0-7-4v-8-4v-servo-separate-power-supply-rc-car-airplane-robot

UBEC CAD: https://grabcad.com/library/henge-ubec-4a-1

2x Servo: https://kstservos.com/products/x08-plus-v6-0-micro-servo-5-3kgf-cm-0-09sec-9-0g-8mm-for-rc-gliders

Power Distribution Board: https://speedyfpv.com/products/drone-power-distribution-board-xt60-3-4s-9-18v-5v-12v-output-pdb?variant=8596736049203&country=US&currency=USD&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&gad_source=1&gad_campaignid=21376024065&gbraid=0AAAAAqe2hTLwYD8IkmY6a0SEM5-ddTCvU&gclid=Cj0KCQjw8c3VBhCsARIsAA_xJ91b9o6USU0LK9LBdmquEBQjtZ6exnOVPERLNiC-Okh-QJNL9f4XGQMaAnQbEALw_wcB

PDB CAD: https://grabcad.com/library/matek-pdb-tx60-1

Battery: https://www.getfpv.com/lumenier-n2o-3500mah-3s-120c-lipo-battery.html?glid=Cj0KCQjw8c3VBhCsARIsAA_xJ92pc9HaFdDk_-xpNVkQ6hIy-5O852-ug0poayFwtlfsRWx3Xk72-P8aAgUxEALw_wcB&afid=aVlOV0hBdmd6THc9&referring_service=google-cpc&utm_source=google&utm_medium=cpc&utm_campaign=DM%20-%20NB%20-%20PMax%20-%20Shop%20-%20No-index%20-%20SM%20-%20ALL%20%7C%20Full%20Funnel&utm_content=pmax_x&utm_keyword=&utm_matchtype=&campaign_id=20799936859&network=x&device=c&gc_id=20799936859&gad_source=1&gad_campaignid=20796067361&gbraid=0AAAAAD8cN5JQHwclR1wGfz4RXDkw2ZIwN&gclid=Cj0KCQjw8c3VBhCsARIsAA_xJ92pc9HaFdDk_-xpNVkQ6hIy-5O852-ug0poayFwtlfsRWx3Xk72-P8aAgUxEALw_wcB

IMU: https://grabcad.com/library/gy-521-mpu6050-accelerometer-and-gyroscope-module-1